Colsonic_16k_I2S_I2C Project Notes

1. Project Overview

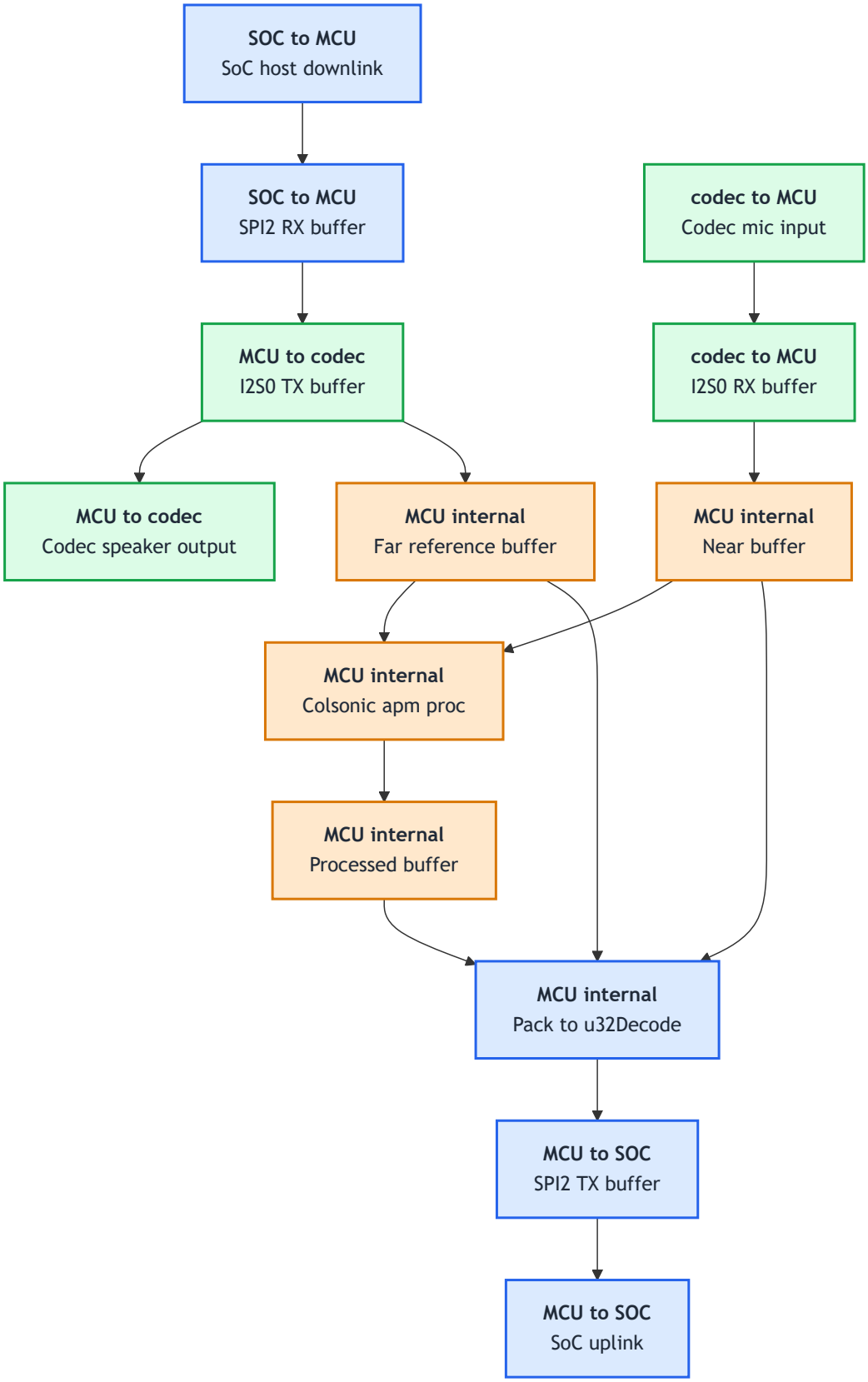
This project is an audio bridge with AEC/NR processing:

- `MCU <-> Codec`： `I2S0` (PD2~PD6)
- `MCU <-> SoC`： `SPI2_I2S` (PA7~PA10)
- Control path: `I2C1` (Codec), `I2C0` (SoC command)

2. Audio Data Flow

Read order:

1. SoC downlink comes in from `SPI2_I2S` .
2. The same downlink is sent to speaker and also used as far reference.
3. Mic capture comes in from `I2S0` as near signal.
4. AEC/NR uses near + far to generate processed voice.
5. Processed result is packed and sent back to SoC uplink.


SOC to MCU
SoC host downlink
SOC to MCU
SPI2 RX buffer
codec to MCU
Codec mic input
MCU to codec
I2S0 TX buffer
codec to MCU
I2S0 RX buffer
MCU to codec
Codec speaker output
MCU internal
Far reference buffer
MCU internal
Near buffer
MCU internal
Colsonic apm proc
MCU internal
Processed buffer
MCU internal
Pack to u32Decode
MCU to SOC
SPI2 TX buffer
MCU to SOC
SoC uplink